

Cloud Infrastructure & Security Standards

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1. Infrastructure Scope & High Availability Targets

This standard defines Service Level Agreements (SLAs), redundancy targets, and security baselines for all cloud resources hosting AI automation workflows, vector stores, and orchestration clusters.

Core automation platforms (n8n, vector endpoints, LLM inference gateways) must maintain 99.99% monthly uptime availability, with scheduled maintenance windows restricted to Sundays 02:00-04:00 UTC.

2. Pinecone Vector Database Configuration & Indexing

The primary production vector repository is hosted on Pinecone serverless clusters across AWS us-east-1 and GCP us-central1 regions.

All production vector indexes must be provisioned with cosine metric indexing and dedicated 3072-dimensional vector spaces to support high-fidelity Google Gemini embeddings. Namespace partitioning must be enforced to isolate development, staging, and corporate knowledge bases.

3. Data Encryption & Key Management (KMS)

All document storage buckets, vector index chunks, and relational metadata ledgers must be encrypted at rest using AES-256 with customer-managed keys rotated annually via Cloud KMS.

All network communication across triggers, webhook listeners, and third-party APIs must require TLS 1.3 encryption with strict cipher suite enforcement.

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4. Disaster Recovery & Backup Objectives (RPO / RTO)

Disaster recovery plans must satisfy a Recovery Point Objective (RPO) of less than 15 minutes and a Recovery Time Objective (RTO) of less than 60 minutes for Tier-1 customer-facing pipelines.

Automated incremental snapshots of vector storage collections and workflow definitions must be replicated cross-region every 6 hours.

5. Rate Limiting, Quotas, and DDoS Mitigation

External webhook endpoints and form triggers must be shielded by Cloudflare Enterprise DDoS mitigation, implementing token-bucket rate limiting capped at 120 requests per minute per IP address.